Explainable Financial Alerts for Retail Investors

Integrating Statistical Anomaly Detection and News–Market Impact Correlation

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Defence

The problem

- Most Americans own stock; many now trade through commission-free apps.
- The US equity market reached **US$62.2 trillion** (end 2024).
- Retail investors face *continuous* price moves and news, **without** the tools of banks and funds.
- The AI spreading through finance is largely **opaque**, a poor fit for someone who must act and live with the result.

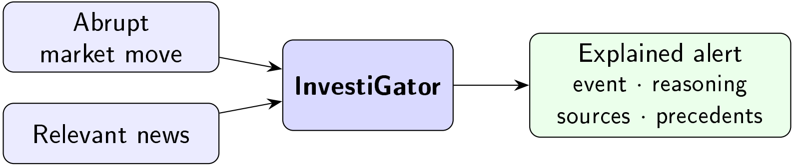
Two distinct challenges

1. Tell a *genuinely* unusual move from ordinary volatility.
2. Read what a news item might *mean*, from how similar events behaved before.

What InvestiGator does (and does not do)

It does *not* predict prices. It explains.

For each abrupt move or relevant news item, it shows *what* was detected, *how*, the *sources*, and which *past events* resemble it, with what happened to prices afterwards.



Research questions

RQ1 Can **transparent** statistical methods detect abrupt moves consistently enough to be useful, while staying explainable?

RQ2 Can a news–market engine retrieve **genuinely analogous** precedents and quantify their impact **without lookahead**, better than trivial baselines?

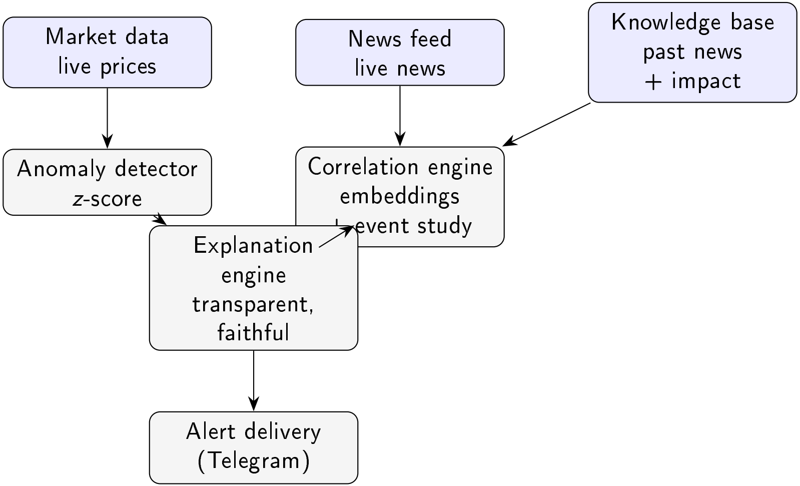
RQ3 Can the explanations be **faithful** to what the system computes, and useful to a retail investor?

RQ4 Can a **supervised model**, trained on historical news and market context, usefully **prioritise** which news deserve an alert, beyond volatility baselines — **never** predicting direction?

Contribution: Artificial Intelligence *Engineering*

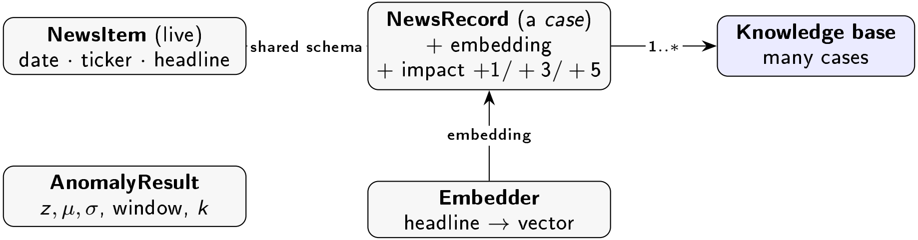
Not a new algorithm: the **disciplined integration, application and critical evaluation** of existing components — plus **one model trained by the author** (materiality triage) — into a working, explainable, reproducible system.

Architecture



Single-responsibility components, common schema; two triggers converge on one explanation step.

The data model — the objects



A *case* = one past headline + what happened to its price. Live **NewsItem** and historical **NewsRecord** share the schema, so a new headline is directly comparable.

The data, at every stage — one real headline

1 • RAW
retrieved
`2018-05-10`
`NVDA`
`"Nvidia Blows`
`Out Q1..."`
+ daily prices

2 • CLEAN
no lookahead
align to 1st trading day
prices → log-returns
drop live summary

3 • REPRESENT AI
model's view
text → SBERT **384-d** vector
context: vol20, mom5, ret
label: material? **yes**

4 • MEASURE
two questions
triage → p(material)
retrieval → precedents
PR-AUC · prec@k

Every value is genuine (Nvidia, 10 May 2018). The "AI" is the **REPRESENT** step: text becomes a 384-dimensional embedding, the event becomes context features, the outcome becomes a label.

Market trigger: transparent by design

Rolling *z*-score, no lookahead:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \quad \text{flag if } |z_t| > k$$

μ_t, σ_t over the window *before* t. Every quantity is exposed to the explanation.

Worked example (real)

TSLA, 24 Oct 2024 (post-earnings):
$\mu = -0.92\%$, $\sigma = 2.73\%$, $r = +19.8\%$
$\Rightarrow z = +7.61$ (flagged at $k = 3$).

A fixed % threshold would treat a calm and a volatile stock alike; the *z*-score does not.

News trigger: retrieve analogous precedents, measure impact

1. Embed the headline (Sentence-BERT).
2. Cosine similarity vs the knowledge base.
3. Return top-k precedents.
4. Measure their impact at $+1/+3/+5$ days, from the event-day close (**no lookahead**).

Case-based reasoning: evidence, not a forecast.

Worked example (real)

Query: "Nvidia demand surges on AI chip orders"

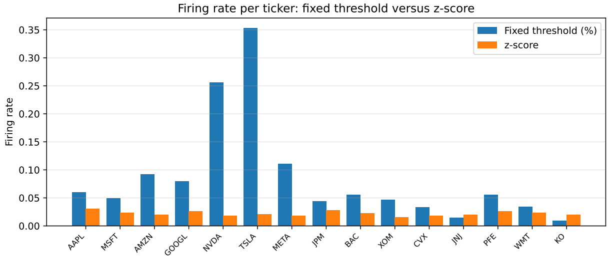
→ NVDA guidance (sim 0.60) +5d +3.5%

→ MSFT cloud-AI (sim 0.38) +5d +10.9%

→ NVDA accelerator (sim 0.38) +5d +4.9%

Cross-ticker match (MSFT); mean +5d ≈ +6.5%.

Result 1: anomaly detection is consistent across the market

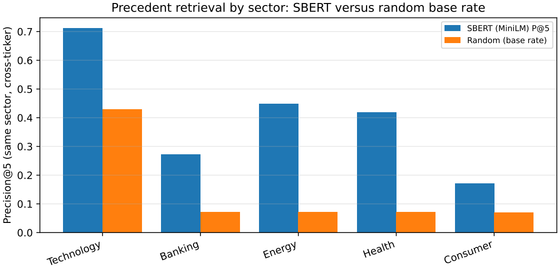


- Firing-rate range across 15 tickers: **0.015** (*z*-score) vs **0.344** (fixed %).
- $> 20\times$ more consistent, a **label-free** argument.
- Vs extreme-move proxy: $F_1 = 0.516$ vs 0.218.
- Window ablation: F_1 0.385 / 0.516 / 0.678 (10/20/60 d).

Result 2: semantic retrieval beats every baseline

Method	P@5	P@10
SBERT (MiniLM)	0.514	0.478
SBERT (MPNet)	0.538	0.506
Lexical	0.346	0.314
Recency	0.126	0.106
Random	0.240	0.240

Cross-ticker, 3,714 headlines, 5 seeds ($\pm \sim 0.01$).



Lift largest where vocabulary is distinctive (energy, health).

Validated at scale on multi-year FNSPID ($\sim$80k headlines): **P@5** 0.595; direction consistency 0.71 sits at the 0.69 chance floor (theme, not direction).

Result 3: explanations are faithful by construction

- Each alert is rendered *directly* from the computed objects.
- So the text **cannot diverge** from the computation.
- Checked by an automated test (alert reproduces each precedent's date/ticker/impact).
- Every alert ends with a no-prediction disclaimer.

Real alert (news trigger)

```
News alert -- NVDA
"Nvidia raises guidance...AI data-centre"
Avg 1-day move: −1.97% (over 5; 3 shown)
- MSFT (0.68) +1d −3.46%
- META (0.68) +1d −2.65%
- GOOGL (0.64) +1d −0.46%
observed past outcome, not a prediction
```

Result 4: learned triage — trained, calibrated, honest

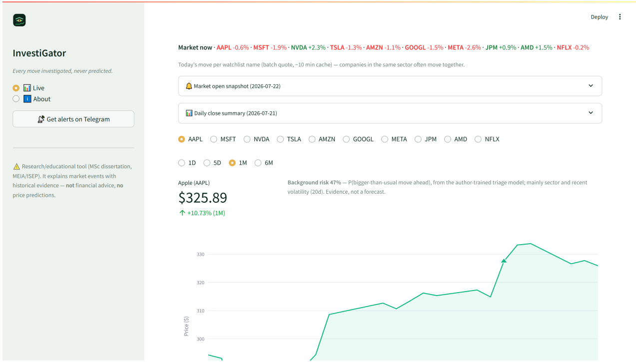
- **Task:** P(abnormal move follows news) — materiality, **never** direction. Labels from our own event study ($|\text{ticker} - \text{SPY}| \geq 2\%$ over $(d, d{+}3]$).
- **Corpus:** FNSPID 2018–2023, 79,753 examples; temporal split by day + embargo; calibration on validation only.
- **Anti-lookahead:** a unit test *mutates future prices* — features unchanged, label changes.

Model	PR-AUC	P@5/day
Always (floor)	0.378	0.163
Volatility LR	**0.542**	**0.632**
Context LR	0.538	0.632
Context+text LR	0.496	0.585
Gradient boosting	0.469	0.551

The honest verdict (pre-committed)

Within a 5-alerts/day budget, triage nearly **quadruples precision** (0.632 vs 0.163) — but **no text model beat volatility**. A follow-up ablation added *five* more cheap context features (market regime, momentum, standardised reaction, downside risk): PR-AUC 0.535, again **no gain**. A *fair* re-test (tuned C, PCA-reduced text, a FinBERT domain encoder) still never beat volatility, so the negative is **robust**, not an under-fit artefact. The signal is market context; reported as it fell.

The product, live



- One page, a tab per watchlist ticker — a genuine screenshot, not a mockup.
- Background risk from the trained triage model, every day, no headline needed.
- Chart annotated with detected events + a history table — the **same** record the Telegram channel received.
- Free tier only: GitHub Actions (cron) + Streamlit Community Cloud.

Built with — free tools and open data, end to end

Data sources & APIs

FNSPID (Hugging Face) | yfinance | Finnhub | RSS feeds | SPY (market proxy)

Machine learning

Sentence-BERT / MiniLM | scikit-learn | PyTorch (CPU) | Platt calibration | ONNX Runtime

Product & delivery

Telegram Bot API | Streamlit | Plotly

Engineering & infrastructure

Python 3.12 | GitHub Actions (cron) | pytest + ruff | joblib

No paid APIs, no GPUs, no always-on server — reproducible on a laptop.

Limitations (stated plainly)

- Relevance is a **same-sector proxy**, not a human judgement.
- Retrieval corpus is a **recent** free-tier window (since validated at scale on FNSPID, P@5 0.595); the *triage* study did use multi-year FNSPID (14/15 tickers — Meta is "FB" in the corpus).
- Displayed precedent impact is a **raw** post-event return (triage labels *are* market-adjusted).
- Similarity is **thematic, not directional**: a cluster can mix up/down moves, so the mean is theme-level *evidence*, not a forecast.
- **No human usefulness study** yet (RQ3 usefulness open).
- By design: **no price prediction, no trading**, so profit metrics are out of scope.

All results are reproducible from versioned scripts (re-run: identical numbers).

Conclusions

- **RQ1 (yes):** transparent *z*-score detects consistently across the market and explains every flag.
- **RQ2 (yes, validated at scale):** semantic retrieval far above baselines (P@5 0.595 on multi-year FNSPID); impact measured without lookahead.
- **RQ3 (faithful yes; useful open):** explanations cannot diverge from the computation; human usefulness is future work.
- **RQ4 (no on text; yes on mechanism):** triage $\approx 4\times$ within-budget precision, but volatility stayed unbeaten — the *second* fair "learned vs simple" test the transparent choice won (first: Isolation Forest vs *z*-score, F_1 0.271 vs 0.530).

Contribution

A working, **explainable, reproducible** financial-alert system for retail investors — existing components plus one author-trained triage model, engineered into a transparent whole, with an honest evaluation.

InvestiGator

Explainable financial alerts for retail investors

Questions welcome.

Anticipated questions

- **Why not predict prices?** Market efficiency (Fama 1970): public news is impounded fast; we *measure* reactions, honestly.
- **Why z-score, not Isolation Forest/GARCH?** Transparency for a low-dimensional signal — and we *tested* it: a causal Isolation Forest on the same information lost (F_1 0.271 vs 0.530).
- **Your trained model lost to the volatility baseline. Failure?** No: the comparison was *pre-committed* and answers RQ4 with evidence; triage still $\approx 4\times$ within-budget precision vs alerting always, and the deployed variant uses exactly the features that carry the signal.
- **How do you know the triage model never sees the future?** A unit test mutates post-event prices and asserts features are unchanged while the label changes; temporal split by unique day with a five-day embargo.
- **Why precision@k and a sector proxy?** Standard ranked-retrieval metric; objective, reproducible; human relevance is future work.
- **Cross-ticker constraint?** An *evaluation* choice to avoid trivial self-matches; the live alert shows the most similar precedents regardless.
- **A positive headline retrieved negative precedents (-1.97%). Misleading?** Similarity matches *theme*, not direction; the mean is theme-level **evidence**, shown with every precedent and a no-prediction disclaimer. Future work: de-duplicate precedents and flag directional disagreement.
- **Is it reproducible?** Yes: fixed seeds, pinned deps, versioned scripts; re-running reproduces every number.